WORKED SOLUTION

STEP I 1998, Question 6

Pure • Trigonometric Recurrence Relations

> **STEP I 1998, Question 6**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/98-s1-q6

The key idea

Every step of the recurrence averages something with a cosine, and the half-angle identity $\frac{1}{2}(1+\cos\theta) = \cos^2\frac{\theta}{2}$ turns that average into a perfect square. The square root in the formula for b_{n+1} then removes the square again. Once we notice that each term contains the previous one as a factor, the general pattern is easy to guess, and checking it is an induction step.

Solution

Throughout, $0 < x < \frac{\pi}{2}$, so every angle $x/2^k$ with $k \geq 1$ lies in $(0, \frac{\pi}{4})$ and its cosine is positive. We take the square root in the definition of b_{n+1} to be the positive one.

Finding a_2 and b_2

Using $1 + \cos\theta = 2\cos^2\frac{\theta}{2}$,

$$a_2 = \tfrac{1}{2}(\cos x + 1) = \cos^2\frac{x}{2}, \qquad b_2 = (a_2 b_1)^{1/2} = \left(\cos^2\frac{x}{2}\right)^{1/2} = \cos\frac{x}{2},$$

where the last step uses $\cos\frac{x}{2} > 0$.

Finding a_3 and b_3

Take out the common factor $\cos\frac{x}{2}$ and then use the half-angle identity again, this time with $\theta = \frac{x}{2}$:

$$a_3 = \tfrac{1}{2}\left(\cos^2\frac{x}{2} + \cos\frac{x}{2}\right) = \cos\frac{x}{2}\cdot\tfrac{1}{2}\left(1+\cos\frac{x}{2}\right) = \cos\frac{x}{2}\cos^2\frac{x}{4}.$$

Then

$$b_3 = (a_3 b_2)^{1/2} = \left(\cos^2\frac{x}{2}\cos^2\frac{x}{4}\right)^{1/2} = \cos\frac{x}{2}\cos\frac{x}{4},$$

since both cosines are positive. These are the two required expressions.

The general pattern

Comparing the terms so far,

$$b_2 = \cos\frac{x}{2}, \quad b_3 = \cos\frac{x}{2}\cos\frac{x}{4}, \qquad a_2 = b_2\cos\frac{x}{2}, \quad a_3 = b_3\cos\frac{x}{4}.$$

Each b gains one more cosine, with the angle halved, and each a is the matching b times its last cosine once more. So we guess that, for $n \geq 2$,

$$b_n = \cos\frac{x}{2}\cos\frac{x}{4}\cdots\cos\frac{x}{2^{n-1}} = \prod_{k=1}^{n-1}\cos\frac{x}{2^k},$$
$$a_n = b_n\cos\frac{x}{2^{n-1}} = \cos\frac{x}{2}\cos\frac{x}{4}\cdots\cos\frac{x}{2^{n-2}}\,\cos^2\frac{x}{2^{n-1}}.$$

Ask yourself What exactly has to be checked to verify that a guess satisfies a recurrence?

We check two things: the guess matches the starting values, and if the formulae hold for a_n and b_n then the recurrence produces the formulae for a_{n+1} and b_{n+1}. Together these show by induction that the formulae hold for every $n \geq 2$.

Starting values. For $n = 2$ the guess gives $b_2 = \cos\frac{x}{2}$ and $a_2 = \cos^2\frac{x}{2}$, which match what we found above.

Inductive step. Suppose the formulae hold for some $n \geq 2$. Write $\theta = x/2^{n-1}$, so that $x/2^n = \theta/2$. Then

$$a_{n+1} = \tfrac{1}{2}(a_n + b_n) = b_n \cdot \tfrac{1}{2}(1+\cos\theta) = b_n\cos^2\frac{\theta}{2}.$$

The guess for b_{n+1} is $b_n\cos\frac{\theta}{2}$ (one more cosine, with angle $x/2^n$), so this says $a_{n+1} = b_{n+1}\cos\frac{x}{2^n}$, which is the guessed form of a_{n+1}. Next,

$$b_{n+1} = (a_{n+1}b_n)^{1/2} = \left(b_n^2\cos^2\frac{\theta}{2}\right)^{1/2} = b_n\cos\frac{x}{2^n} = \prod_{k=1}^{n}\cos\frac{x}{2^k},$$

taking the positive root, which is legitimate because b_n and $\cos\frac{x}{2^n}$ are both positive. That is the guessed form of b_{n+1}, so both equations of the recurrence are satisfied and the formulae hold for all $n \geq 2$.

In fact, with the empty product equal to 1, the formulae also give $b_1 = 1$ and $a_1 = \cos x$, so they hold from $n = 1$.

Checks

- Take $x = \frac{\pi}{3}$. Running the recurrence directly gives $a_3 \approx 0.808013$ and $b_3 \approx 0.836516$, and $\cos\frac{\pi}{6}\cos^2\frac{\pi}{12}$ and $\cos\frac{\pi}{6}\cos\frac{\pi}{12}$ give the same values.
- For 2000 random x in $(0, \frac{\pi}{2})$ we compared the recurrence with the product formulae up to $n = 24$: they agreed to within 10^{-15}.

- As $x \to 0$, every cosine tends to 1, and so do a_1 and b_1. All the terms tend to 1, as they should.

Common slips

- Writing $\sqrt{\cos^2\theta} = \cos\theta$ without saying why $\cos\theta > 0$. Here it holds because every angle lies in $(0, \frac{\pi}{4})$.
- Guessing the pattern from $n = 2$ and 3 and stopping there. The question asks us to verify the recurrence, so the inductive step is needed.

Beyond STEP

Multiplying b_n by $\sin\frac{x}{2^{n-1}}$ and using $2\sin\phi\cos\phi = \sin 2\phi$ repeatedly gives $b_n = \dfrac{\sin x}{2^{n-1}\sin(x/2^{n-1})}$. This tends to $\dfrac{\sin x}{x}$ as $n \to \infty$ (with $x = 1$ the recurrence gives 0.841471 after a few dozen steps). Taking $x = \frac{\pi}{2}$ gives Viète's infinite product for $2/\pi$. This averaging-then-geometric-mean scheme is a cousin of Gauss's arithmetic–geometric mean, and makes a good first example of proving that a sequence converges and finding its limit. For a careful treatment of limits of sequences, see Abbott, *Understanding Analysis*.

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